

The University of Chicago Data Science Institute AI for Climate (AICE) Initiative advances climate forecasting research to provide more accurate and actionable predictions for agricultural stakeholders, namely farmers and policymakers. To support these efforts, AICE partnered with the Data Science Clinic on two projects: (1) completing development of the “monsoonbench” repository, a critical supplementary framework for reproducing the analysis conducted in an AICE research paper¹ focused on monsoon onset over India, and (2) contributing to the Rainy Onset Metrics Package (ROMP), a codebase for evaluating AI models on monsoon onset prediction tasks in Ethiopia, and eventually worldwide.

The team transformed the monsoonbench repository into a clean, easy-to-use pipeline for reproducing the exact figures from the research paper directly from raw data. The repository was subsequently published as a PyPI package, enabling researchers to apply the benchmarking systems developed by the team to future forecasting models.

The team also extended ROMP with new error metrics, visualizations, and benchmarking methods. One notable addition was Contiguous Rain Area (CRA) verification, a technique that evaluates forecasts as spatial objects rather than independent grid cells. The figure below demonstrates the value of CRA verification. While the AI model’s forecasted rainfall (center) captures rainfall magnitudes similar to the observed data (left), the forecast is spatially displaced. Simply shifting the forecast two grid cells westward, as indicated by the arrow, greatly improves spatial alignment and overall forecast accuracy (right). Ongoing development will continue in collaboration with the Ethiopian Meteorological Institute, to which the team recently presented these additions.

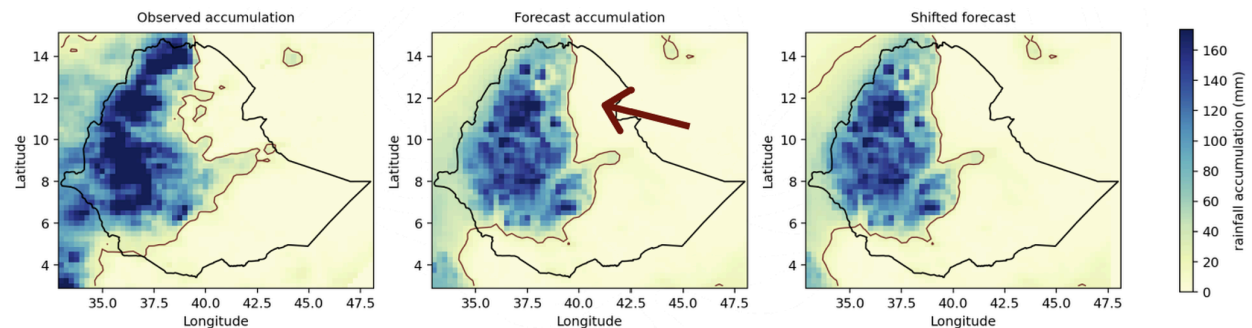


Figure 1: Observed vs. Predicted vs. Shifted Rainfall across Ethiopia for the AIFS forecasting model (2016).

¹ <https://arxiv.org/abs/2602.03767>